

Chapter 2

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet: "Science Superpower Challenge" for Chapter 2: Foundations of Science Purpose: This worksheet is designed to engage high school students with the core concepts of Physics, Chemistry, Biology, the Scientific Method, and hands-on experiments from Chapter 2. Through a mix of creative tasks, quick questions, and mini-challenges, students will explore science in a fun, interactive way while reinforcing key ideas. **Materials Needed:** Pen/pencil, notebook, household items for mini-experiments (string, vinegar, baking soda, etc.), phone/camera (optional for sharing)

Mission 1: Physics Power-Up – Motion, Forces, and Energy (15 minutes)

Task 1: Quick Brainstorm (5 minutes) Think about your favourite sport or activity (like skateboarding, basketball, or dancing). Pick one moment—like a jump or a throw—and write down 1 way motion, 1 way forces, and 1 way energy are involved. Example: In basketball, motion is the ball flying to the hoop, force is gravity pulling it down, and energy is the kinetic energy as it moves.

- My Activity: _____
- Motion: _____
- Force: _____
- Energy: _____

Task 2: Mini-Challenge – "Ramp Racer" (10 minutes) Using a book and a small toy car (or a rolled-up paper ball), make a ramp. Slide your "racer" down and observe how fast it goes. Change the ramp's angle (steeper or flatter) and note what happens to the speed. Write 1 sentence about why you think the speed changed (hint: think forces or energy).

- Observation: _____

Mission 2: Chemistry Quest – Elements, Compounds, and Reactions (15 minutes)

Task 1: Mix & Match (5 minutes) Draw lines to match these everyday items to their chemical makeup or reaction:

- Soda fizz → Carbon dioxide gas from a reaction
- Table salt → Compound of sodium and chlorine (NaCl)
- Pencil lead → Element carbon (C)
- Water → Compound of hydrogen and oxygen (H₂O)

Task 2: Mini-Challenge – "Fizz Wizard" (10 minutes) Mix 1 tablespoon of vinegar with ½ teaspoon of baking

soda in a small cup (over a sink!). Watch the reaction. Add a drop of food coloring if you've got it for extra coolness. Write down 2 things you noticed (like bubbles or sound) and 1 reason this is a chemical reaction.

- Noticed 1: _____
- Noticed 2: _____
- Why it's a reaction: _____

Mission 3: Biology Breakthrough – Cells, Ecosystems, and Human Body Systems (15 minutes)

Task 1: Life Detective (5 minutes) Look around your room or outside a window for 1 minute. List 3 living things you see (plants, animals, bugs) and guess how they might connect in an ecosystem (like who eats who or helps who).

- Living Thing 1: _____
- Living Thing 2: _____
- Living Thing 3: _____
- Connection Guess: _____

Task 2: Mini-Challenge – “Breath Blast” (10 minutes) Breathe out through a straw onto a small mirror or phone screen for 10 seconds. See the fog? That's from your respiratory system! Now do 10 jumping jacks and try again. Is the fog bigger? Write 1 sentence about why exercise might change the fog.

- Observation: _____

Mission 4: Scientific Method Mystery – Think Like a Scientist (10 minutes)

Task: Solve a Daily Dilemma Pick one weird thing you've noticed lately (like “Why does my phone battery drain faster some days?”). Follow the scientific method steps below to start cracking the case!

- **Question:** What do you want to figure out?
- **Hypothesis:** Make a guess (if-then style).
- **Quick Test Idea:** How could you test this at home?

Mission 5: Super Science Showdown – Pick Your Power (15 minutes)

Task: Choose 1 Experiment to Rule Them All! Pick one of the hands-on activities from the chapter (Pendulum Power, Fizzy Fun, or Breath Test) that you haven't done yet—or redo a favourite. Do the experiment and answer these:

- Which did you pick? _____
- What was the coolest part? _____
- How does this connect to a science idea from the chapter (like forces or reactions)?

Bonus: Snap a pic or record a 15-second video of your experiment. Share it with your class group chat or tag a friend to challenge them!

Final Challenge: Science Superhero Name & Reflection (5 minutes)

Task 1: Create Your Alias Based on your favourite part of this worksheet, make up a science superhero name for yourself (like “Force Falcon” or “Reaction Rebel”). Write it here:

- My Superhero Name: _____

Task 2: Reflect & Share Write 1 sentence about the most surprising or fun thing you learned from Chapter 2 or this worksheet.

- My Big Takeaway: _____
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